CHENXI FENG

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SUMMARY

Geospatial data scientist with 3+ years of hands-on experience developing end-to-end data pipelines for large-scale environmental datasets, including hyperspectral imageries, climate reanalysis, and multi-source meta-datasets. Experienced in building automated retrieval and data-model fusion systems in Python, with applications in atmospheric methane detection and carbon cycle modeling. Seeking to apply geospatial scientific computing and environmental data analytics to sustainability measurement and environmental risk quantification

SKILLS

- **Core Competencies**: Data analysis and visualization, Modeling, Machine Learning
- **Programming**: Python, MATLAB, R, FORTRAN
- **Tools & Software**: ArcGIS, ENVI, Google Earth Engine, Git, Linux/Bash, Microsoft office
- **Languages**: English (fluent), Mandarin Chinese (native)

EDUCATION

Cornell University Ithaca, NY, United States | 09/2024 – Present

Ph.D. in Atmospheric Science

- **Focus**: AI-guided data-model fusion; Solar geoengineering; Land-based carbon sequestration; Climate sensitivity and resilience
- **Relevant Coursework**: Data-model fusion in research; Climate dynamics

Nanjing University Nanjing, Jiangsu, China | 09/2020 – 06/2024

B.S. in Atmospheric Science | GPA: 3.7/4

- **Focus**: Retrieval algorithm development; Remote sensing data processing; Greenhouse gas emission detection
- **Relevant Coursework**: Data assimilation with big data; Atmospheric measurement techniques; Earth observations
- **Honors & Awards**: First Prize Outstanding Thesis Award (2024); Academic Excellence Scholarship (2023)

DATA SCIENCE EXPERIENCE

Cornell University Ithaca, NY, United States | 09/2024 – Present

Graduate Research Assistant

- **Data Integration & Modeling**: Integrated heterogeneous multi-source datasets into a unified global meta-database and developed a data-model fusion workflow for land carbon cycle research
- **Data Upscaling & Machine Learning**: Built automated spatial upscaling workflows to transform sparse, multi-site observations into high-resolution global gridded data products using biogeochemistry-informed neural network architectures

ALGORITHM DEVELOPMENT EXPERIENCE

California Institute of Technology Pasadena, CA, United States | 06/2023 – 01/2024

Undergraduate Research Intern

- **Quality Assurance & Bias Calibration**: Developed and deployed an effective aerosol-calibrated atmospheric methane retrieval algorithm to eliminate inversion biases for airborne and satellite hyperspectral imageries

Nanjing University Nanjing, Jiangsu, China | 02/2022 – 07/2024

Undergraduate Research Assistant

- **Image Processing & Feature Engineering**: Developed an image processing pipeline for hyperspectral remote sensing observations, extracting and fusing multi-channel spectral features to accurately detect and map point-source methane plumes

SELECTED PUBLICATIONS

(*, co-fisrt author)

- **Feng, C.**, Zhang, S., Qin, S., Xu, X. et al., 2026. Global convergence in temperature responses of microbial respiratory soil organic carbon loss. *In prep.*
- Li, F.*, **Feng, C.***, Juan, J. R., Guanter, L. et al., 2026. An optimized methane retrieval approach based on morphological fusion for mapping methane point emissions from spaceborne imaging spectrometry. Remote Sensing of Environment. 10.1016/j.rse.2026.115718.
- **Feng, C.**, Chen, S., Zeng, Z.-C., Luo, Y., Natraj, V., & Yung, Y. L., 2024. Aerosol-Calibrated matched filter method for retrievals of methane point source emissions over the Los Angeles Basin. Earth and Space Science. 10.1029/2024EA003519.